EMC404S Mechanisms Explained

Why the patient looks like that, and why the treatment works — for the viva voce

How to use this section

Examiners at a viva rarely stop at "what do you give?". The next question is almost always "**why** does that work?" or "**why** does the patient present like that?". Every mechanism below is written as a chain of four links, and if you can say all four links, you can answer any follow-up question:

1. **What the poison or condition does in the body** (the mechanism).
2. **What you therefore see** (each sign is explained by the mechanism, so you never need to memorise signs as a random list).
3. **Why the treatment works** (each treatment reverses or bypasses one link in the chain).
4. **The sentence to say in the viva.**

Treatments marked *(hospital)* are not within prehospital scope in South Africa, but you must be able to explain them, because they explain why you transport and what you hand over.

Part A — Behavioural emergencies: the mechanisms

A1. Why a "psychiatric" patient may really have a medical cause

The brain is the organ with the highest demand for **oxygen and glucose** and the smallest reserves of either, so any condition that deprives it of oxygen, glucose, normal electrolytes, normal temperature or a normal neurotransmitter balance shows up first as a change in **behaviour** — confusion, agitation, aggression or psychosis — long before other organs fail. This is why the CDST instructs providers not to assume that aggression is due to a psychiatric condition, and to check the ABCs, oxygen saturation, glucose and other reversible causes first (HPCSA PBEC, 2025).

- **Hypoglycaemia** → the neurons run out of fuel (**neuroglycopenia**) → confusion, aggression, slurred speech and focal signs that look like intoxication or a stroke; at the same time, the body releases adrenaline in response to the low glucose, which produces sweating, tremor and tachycardia. **Treatment logic**: glucose restores the fuel directly, so the behaviour normalises within minutes. **Say**: "*Every agitated patient gets a glucose, because hypoglycaemia is the commonest reversible cause of abnormal behaviour and it kills if it is missed.*"
- **Hypoxia** → the neurons switch to anaerobic metabolism and fail → restlessness and agitation are often the **first** sign of hypoxia, before cyanosis appears. **Treatment logic**: oxygen when SpO_2 is below 92% (HPCSA PBEC, 2025).
- **Delirium** → an acute disturbance in attention and awareness, which fluctuates, caused by an underlying medical condition, substance or withdrawal (APA, 2022). The leading explanations are an imbalance of neurotransmitters — in particular a **relative deficiency of acetylcholine and an excess of dopamine** — together with inflammation and stress responses acting on a vulnerable brain (Inouye, Westendorp and Saczynski, 2014). This mechanism explains three viva points:
    1. **Anticholinergic drugs** (for example antihistamines, tricyclic antidepressants, *Datura*) cause delirium, because they further reduce acetylcholine activity.
    2. **Older people with dementia** become delirious easily, because they have fewer cholinergic neurons in reserve.
    3. **Dopamine-blocking drugs** (haloperidol) can calm hyperactive delirium, but they do not treat its cause.

A2. Alcohol withdrawal — why it causes seizures and delirium tremens

Alcohol **enhances GABA-A** (the brain's main inhibitory, "brake" system) and **inhibits NMDA glutamate** receptors (the main excitatory, "accelerator" system). With chronic heavy drinking, the brain adapts to keep functioning: it **reduces GABA-A function and increases NMDA receptor activity**. When the person suddenly stops drinking, the alcohol "brake" disappears, but the adapted brain still has a weak brake and a strong accelerator, which leads to **unopposed excitation** (Schuckit, 2014; Rang and Dale — Ritter et al., 2023).

- **What you see**, in time order: tremor, anxiety, sweating, tachycardia and hypertension (from about 6–24 hours); withdrawal **seizures** (typically within about 48 hours); then

delirium tremens (from about 48–96 hours) — confusion, hallucinations, autonomic hyperactivity and hyperthermia — which can be fatal (Schuckit, 2014).

- **Treatment logic**: **benzodiazepines** act on the same GABA-A receptor as alcohol, so they **replace the missing brake**. This is why benzodiazepines, not antipsychotics, are the treatment of choice, and antipsychotics alone may lower the seizure threshold. **Thiamine** is given because people with chronic alcohol use and malnutrition are thiamine-deficient (see C12).
- **Say**: "*Withdrawal is the brain's brake being suddenly removed after it has adapted to alcohol, so I give a benzodiazepine, which acts on the same GABA-A receptor, plus thiamine.*"

A3. Acute behavioural disturbance (ABD) and why restraint can kill

ABD is a state of extreme agitation with **abnormal physiology**, often driven by stimulant drugs, psychosis or both. The RCEM (2025) explains why continued struggling against restraint is dangerous: "continued exertion under restraint can contribute to poor outcomes (likely due to increasing **catecholamine** levels, worsening **hyperthermia**, and **metabolic acidosis**)".

The chain that leads to sudden death can be explained as follows:

1. Intense, prolonged muscle activity and a massive **catecholamine surge** → tachycardia, hypertension and a very high oxygen demand.
2. The muscles work anaerobically → **lactic acid** accumulates → **metabolic acidosis**. Heat production rises, and sweating and vasoconstriction may not keep up → **hyperthermia**.
3. The body compensates for the acidosis by **breathing faster and deeper** to blow off CO_2 (respiratory compensation).
4. Anything that **restricts breathing** removes that compensation. Examples are prone positioning with weight on the back or chest, a restraint hold across the torso or neck, or over-sedation with respiratory depression. The acidosis worsens very quickly, potassium shifts out of cells, and the catecholamine-sensitised heart develops an **arrhythmia or arrest**.

Treatment logic — each CDST step breaks one link:

- **Verbal de-escalation** first reduces the struggle (less exertion means less acid and heat).
- If physical restraint is necessary, use **enough people** (the CDST uses five people plus one person to coordinate) so that the restraint is brief and controlled, and **never** put weight on the chest, back or neck, which keeps the respiratory compensation intact.
- **Early sedation** (midazolam IM) stops the muscle activity that produces acid and heat.
- **Observations every 10 minutes** (SpO_2, respiratory rate, level of consciousness, temperature) detect deterioration early, followed by cooling and transport (HPCSA PBEC, 2025).

Say: "*The danger in ABD is not the aggression itself but the physiology: catecholamines, acidosis and hyperthermia. Restraint that restricts breathing removes the patient's respiratory*

compensation, so I keep restraint brief, never prone with weight, sedate early to stop the exertion, and monitor every ten minutes."

A4. Sedation drugs — how each one works and why it is chosen

Midazolam and lorazepam (benzodiazepines)

- **Mechanism**: they bind to the **benzodiazepine site on the GABA-A receptor** and increase the **frequency** with which its chloride channel opens when GABA binds. More chloride enters the neuron, so the neuron is **hyperpolarised** (harder to fire) and the brain is calmed (Ritter et al., 2023).
- **Why they work in agitation**: they reduce anxiety and arousal, sedate, and are **anticonvulsant** — useful in stimulant toxicity, withdrawal and unknown ingestions.
- **Why midazolam suits IM use**: it is water-soluble in its ampoule, so it is absorbed reliably from muscle (Ritter et al., 2023). This matters when you cannot safely obtain IV access in a fighting patient.
- **The main danger is respiratory depression**, which is much greater when the patient has also taken **alcohol, opioids or other sedatives**. All of these depress the brain through overlapping systems, so the effects add together or multiply.
- **Why the dose is halved in people over 60**: older people clear the drug more slowly and their brains are more sensitive, so they become over-sedated, fall and aspirate. This is why the CDST reduces the dose to 0.1 mg/kg if the patient is over 60 years old (HPCSA PBEC, 2025), and why benzodiazepines are on the Beers Criteria list (American Geriatrics Society, 2023).

Ketamine (ANT/ECP in the CDST)

- **Mechanism**: it is an **NMDA receptor antagonist** — it blocks the glutamate accelerator — producing a **dissociative** state in which the patient is disconnected from the surroundings, with strong analgesia (Ritter et al., 2023).
- **Why it is useful**: it has a fast onset after IM injection, and **airway reflexes and respiratory drive are relatively preserved** compared with other sedatives. It also **stimulates the sympathetic nervous system**, so blood pressure is usually maintained.
- **Its dangers follow from the same mechanism**: sympathetic stimulation raises the heart rate and blood pressure; it causes **hypersalivation** and occasionally **laryngospasm**; and it can cause **emergence reactions** (frightening dreams and hallucinations as it wears off) (Ritter et al., 2023).

Haloperidol and other antipsychotics (mainly used in hospital, according to the guidelines you were given)

- **Mechanism**: they block **dopamine D2 receptors**. In the **mesolimbic pathway**, this reduces psychotic symptoms; this is the basis of the dopamine hypothesis of schizophrenia (McCutcheon, Reis Marques and Howes, 2020).
- **Side effects follow from where else D2 is blocked**:

  - In the **nigrostriatal pathway** (movement control), D2 blockade causes **acute dystonia** (for example oculogyric crisis and torticollis), akathisia and parkinsonism.
  - Rarely, it causes **neuroleptic malignant syndrome** (rigidity, hyperthermia and autonomic instability).
  - **Potassium channel blockade** in the heart causes **QT prolongation and torsade de pointes** (Ritter et al., 2023).
- **Why they are avoided in Parkinson's disease and Lewy body dementia**: these patients already lack dopamine, so D2 blockade severely worsens them (RCEM, 2025).

Part B — The toxidromes: why each sign appears

A toxidrome makes sense once you know **which receptor controls which organ**. The key fact that most students miss is that **sweat glands are supplied by sympathetic nerves but use acetylcholine acting on muscarinic receptors** (Ritter et al., 2023). This single fact explains why the two "big pupil" toxidromes differ in skin.

Toxidrome	Mechanism	Why the pupils look like that	Why the skin looks like that	Other signs explained
Sympathomimetic (cocaine, amphetamines, "tik")	Too much **noradrenaline and dopamine** (release or blocked reuptake)	α_1 receptors contract the radial (dilator) muscle of the iris → **big pupils**	The sympathetic nervous system drives the sweat glands (via muscarinic receptors, which still work) → **wet, sweaty skin**	Tachycardia, hypertension, agitation, hyperthermia from muscle activity and vasoconstriction
Anticholinergic (*Datura*, antihistamines, TCAs)	**Muscarinic receptors are blocked**	The pupil sphincter is muscarinic; when it is blocked, the dilator pulls unopposed → **big pupils**; the ciliary muscle is paralysed → **blurred vision** ("blind as a bat")	The sweat glands are muscarinic, so they are blocked → **dry skin** ("dry as a bone"); the body cannot sweat, so it overheats ("hot as a hare") and dilates skin vessels to lose heat ("red as a beet")	The bladder detrusor is muscarinic → retention ("full as a flask"); gut → absent bowel sounds; brain → delirium ("mad as a hatter"); heart (vagal brake blocked) → tachycardia
Opioid	**μ-opioid receptor** agonism	Stimulation of the **Edinger–Westphal nucleus** → **pinpoint pupils**	Normal	The brainstem becomes less responsive to CO_2 and hypoxia → **respiratory depression** (the cause of death)
Cholinergic (organophosphates, carbamates)	**Acetylcholinesterase is inhibited**, so acetylcholine accumulates at every synapse that uses it	Muscarinic overstimulation of the sphincter → **pinpoint pupils**	Muscarinic overstimulation of the sweat glands → **drenched skin**	Muscarinic → DUMBELS and **bronchorrhoea and bronchospasm**; nicotinic (muscle) →

Toxidrome	Mechanism	Why the pupils look like that	Why the skin looks like that	Other signs explained
				fasciculations, then weakness and paralysis; CNS → seizures and coma (Eddleston et al., 2008)
Sedative-hypnotic (benzodiazepines, alcohol)	Enhanced **GABA-A** activity	Usually normal	Normal	Global CNS depression with relatively preserved vital signs in pure benzodiazepine overdose
Serotonin toxicity	Excess **serotonin** (5-HT, especially 5-HT2A)	Big pupils	Wet (sweaty)	**Clonus, hyperreflexia**, tremor, hyperthermia, diarrhoea; rapid onset (hours) after a serotonergic drug is added or increased (Boyer and Shannon, 2005)

The discriminating rule and its reason: "Big and wet = sympathomimetic (the sweat glands still work); big and dry = anticholinergic (the sweat glands are blocked); small and wet = cholinergic; small and normal skin with slow breathing = opioid."

Serotonin toxicity vs neuroleptic malignant syndrome (NMS):

- **Serotonin toxicity** is a state of **excess serotonin**, so the nervous system is "**switched on**": onset within hours, **clonus and hyperreflexia**, and big pupils (Boyer and Shannon, 2005).
- **NMS** is a state of **dopamine blockade**, so movement is "**switched off**": onset over days, "lead-pipe" **rigidity**, slow reflexes, and normal pupils (Boyer and Shannon, 2005).

Say: "*The toxidrome follows the receptor: sweat glands are muscarinic, so a patient with big pupils and dry skin is anticholinergic, while big pupils with sweating is sympathomimetic.*"

Part C — Specific poisons: mechanism → signs → treatment

C1. Paracetamol — "the quiet killer"

- **Mechanism**: about 90% of a normal dose is made safe by conjugation (glucuronidation and sulphation), and 5–10% is converted by **cytochrome P450 (CYP2E1)** into the highly reactive **NAPQI**, which is immediately neutralised by **glutathione**. In overdose, the safe conjugation pathways are **saturated**, so much more NAPQI is made. Once **glutathione stores fall below about 30%**, free NAPQI binds to proteins inside the liver cells and destroys them, causing **centrilobular (zone 3) necrosis** — the zone where CYP2E1 is most concentrated (Tintinalli et al., 2020).
- **Why the patient looks well for 24 hours**: the liver cells are still dying silently. Symptoms (right upper quadrant pain, jaundice, bleeding, encephalopathy and hypoglycaemia) only appear once enough hepatocytes have been destroyed, usually after 24–72 hours. **A well-looking patient is not a safe patient.**
- **Why some people are at higher risk at lower doses**: **chronic alcohol use and enzyme-inducing drugs** (some anti-TB drugs and anticonvulsants) **increase CYP2E1**, so more NAPQI is produced. **Malnutrition** means less glutathione in reserve (Hüser et al., 2025).
- **Why NAC works**: **N-acetylcysteine** is a **glutathione precursor**, so it **replenishes glutathione** and NAPQI is neutralised again. It is most effective **within 8 hours** of ingestion, before significant liver cell death has occurred (Tintinalli et al., 2020). This is why prehospital delay costs livers.
- **Why the blood level is taken at 4 hours or later**: before 4 hours, absorption is incomplete, so an early level underestimates the dose (Chiew et al., 2020).
- **Say**: "*In overdose, glutathione is depleted and the toxic metabolite NAPQI destroys zone 3 hepatocytes; the patient looks well for a day, so I transport every paracetamol overdose, because NAC replaces glutathione and works best within eight hours.*"

C2. Tricyclic antidepressants (TCAs) — "talks, fits, arrests"

- **Mechanism**: one drug with six actions (Hüser et al., 2025; Tintinalli et al., 2020):
  1. **Fast sodium channel blockade** in the heart — the killer.
  2. **Muscarinic blockade** (anticholinergic).
  3. **α_1 blockade**.
  4. Blocked reuptake of noradrenaline and serotonin.
  5. **GABA-A antagonism**.
  6. **Potassium channel blockade**.
- **Signs explained**:
  - **Sodium channels** are responsible for the rapid upstroke of the ventricular action potential, so blocking them slows conduction → **wide QRS**, terminal R wave in aVR, VT and VF, and a weaker contraction → hypotension.
  - **Muscarinic blockade** → the anticholinergic picture (tachycardia, big pupils, dry skin, retention). It also slows the gut, so absorption continues for longer.
  - **α_1 blockade** → vasodilatation → hypotension.
  - **GABA-A antagonism** → **seizures**.

  - The QRS width predicts what happens next: **QRS above 100 ms is associated with seizures, and QRS above 160 ms with ventricular arrhythmias** (Hüser et al., 2025).
- **Why they deteriorate so fast**: absorption is rapid, and the drug acts on the brain and heart simultaneously, so a patient who is talking can be fitting and in VT within minutes.
- **Why sodium bicarbonate works**: two effects (Hüser et al., 2025).
  1. The **sodium load** increases the sodium gradient across the cell membrane, so more sodium flows through the channels that remain unblocked, and the QRS narrows.
  2. **Alkalinisation** (target blood pH about 7.45–7.55) reduces the binding of the TCA to the sodium channel.
  - This is also why the CDST says that if the patient is intubated, you **hyperventilate to an $ETCO_2$ of 20–25 mmHg**: you are deliberately producing a respiratory alkalosis (HPCSA PBEC, 2025).
- **Why no flumazenil and no physostigmine**: TCA toxicity is held in check partly by any co-ingested benzodiazepine, and flumazenil can unmask **seizures**. Physostigmine increases acetylcholine in a heart whose conduction is already blocked, risking **bradycardia and asystole**, so it is contraindicated when the QRS is wide (Hüser et al., 2025).
- **Say**: "*TCAs block cardiac sodium channels, which widens the QRS and causes VT; sodium bicarbonate overcomes the block with a sodium load and alkalinisation reduces drug binding, so a QRS over 100 ms means bicarbonate now.*"

C3. Salicylates (aspirin) — "breathing fast for two reasons"

- **Mechanism and signs** (Tintinalli et al., 2020):
  1. Salicylate **directly stimulates the respiratory centre** → hyperventilation → **respiratory alkalosis** (the first finding in adults).
  2. It **uncouples oxidative phosphorylation**: the mitochondria burn fuel, but the energy is released as **heat** instead of being captured as ATP → **hyperthermia**, sweating, increased CO_2 production and anaerobic metabolism → **high anion gap metabolic acidosis**. The adult therefore shows a **mixed respiratory alkalosis and metabolic acidosis**.
  3. Salicylate is toxic to the inner ear → **tinnitus** and deafness.
  4. Increased capillary permeability → non-cardiogenic **pulmonary and cerebral oedema**.
- **The most important mechanism for practice — acid makes it worse**: salicylate is a weak acid. When the blood becomes **more acidic**, more salicylate exists in the **non-ionised, lipid-soluble** form, which **crosses into the brain**, and CNS salicylate toxicity is the main cause of death (Tintinalli et al., 2020).
- **Why you must never let a salicylate patient hypoventilate**: if you sedate or intubate them and ventilate at a "normal" rate, you take away their hyperventilation. The blood becomes acidic, salicylate moves into the brain, and the patient can arrest. **Match their high minute ventilation.**

- **Why alkalinisation helps** *(hospital)*: alkaline blood keeps salicylate **ionised**, trapping it outside the brain. Alkaline urine traps ionised salicylate in the urine so that it is excreted (**ion trapping**). **Dialysis** is used in severe poisoning (Juurlink et al., 2015).
- **Say**: "*Salicylate uncouples oxidative phosphorylation, causing hyperthermia and metabolic acidosis, and acidosis drives more salicylate into the brain, so never let them hypoventilate.*"

C4. Iron — "the dangerous quiet phase"

- **Mechanism** (Tintinalli et al., 2020):
    1. **Directly corrosive** to the stomach and gut lining → vomiting, haematemesis and bloody diarrhoea, with fluid and blood loss.
    2. Once the iron-carrying protein **transferrin is saturated**, **free iron** enters cells, **poisons mitochondrial oxidative phosphorylation** and generates **free radicals**. This causes **lactic acidosis**, shock (hypovolaemic, distributive and cardiogenic), coagulopathy and **liver necrosis** — the liver is hit first because it receives the iron from the gut via the portal vein.
- **Why there is a "latent phase"** (6–24 hours): the gut symptoms settle, so the patient seems better, but the free iron is still destroying cells. This **false reassurance** is a viva favourite.
- **Why no charcoal**: iron is a small, ionised metal and is **not adsorbed by charcoal** (HPCSA PBEC, 2025). **Whole bowel irrigation** is used in hospital instead.
- **Why deferoxamine works** *(hospital)*: it is a **chelator** — it binds free iron to form **ferrioxamine**, which is excreted by the kidneys and may turn the urine a reddish ("vin rosé") colour (Tintinalli et al., 2020).
- **Say**: "*Iron is corrosive to the gut and then, as free iron, poisons the mitochondria; the latent phase is false reassurance, charcoal does not bind it, so I transport.*"

C5. Lithium — "the kidney decides"

- **Mechanism**: lithium has a **narrow therapeutic index** and is excreted almost entirely by the **kidney**, which handles it **like sodium**. When the patient is dehydrated or sodium-depleted (vomiting, diarrhoea, diuretics or NSAIDs), the kidney holds on to sodium and **holds on to lithium with it**, so levels rise. Lithium also causes **nephrogenic diabetes insipidus** (polyuria), which causes more dehydration, so the problem feeds itself. Lithium enters the brain slowly and leaves it slowly, which is why **chronic toxicity** (slow accumulation) causes more neurotoxicity than the blood level suggests (Tintinalli et al., 2020; Decker et al., 2015).
- **Signs**: GI upset first (acute); then **coarse tremor**, ataxia, confusion, hyperreflexia, seizures and coma (neurotoxicity).
- **Why there is no charcoal**: lithium is a small, ionised metal and is not adsorbed (HPCSA PBEC, 2025).
- **Why saline helps**: restoring sodium and volume lets the kidney excrete lithium again.
- **Why dialysis works so well** *(hospital)*: lithium is a small, water-soluble molecule with a small volume of distribution that is not bound to protein, so it is easily removed by haemodialysis (Decker et al., 2015).

- **Say**: "*The kidney treats lithium like sodium, so dehydration raises lithium levels; it isn't bound by charcoal, and the treatment is fluids and dialysis.*"

C6. Sulphonylureas — "one tablet, one toddler, a long night"

- **Mechanism**: sulphonylureas **close the ATP-sensitive potassium channels** on the pancreatic β-cell. The cell depolarises, calcium enters, and **insulin is released regardless of the blood glucose level** (Ritter et al., 2023).
- **Signs**: **hypoglycaemia** — adrenergic (sweating, tremor, tachycardia) and neuroglycopenic (confusion, aggression, seizures and coma) — that is **delayed** (up to 8–16 hours or more), **prolonged** (24–72 hours) and **recurrent** (Tintinalli et al., 2020).
- **Why it recurs after you give dextrose**: the β-cell is still being driven by the drug, so **glucose itself stimulates even more insulin release**. The glucose rises, insulin surges, and the patient becomes hypoglycaemic again.
- **Why octreotide works** *(hospital)*: it is a somatostatin analogue that **inhibits insulin release** from the β-cell, breaking this cycle (Tintinalli et al., 2020).
- **Why every child is transported**: a single tablet can cause serious hypoglycaemia in a toddler, and it may only appear many hours later.
- **Say**: "*Sulphonylureas force insulin release, and the dextrose I give stimulates even more, so the hypoglycaemia is delayed and recurrent; I transport every exposure.*"

C7. Organophosphates and carbamates (including aldicarb, "Two-step")

- **Mechanism**: organophosphates **phosphorylate acetylcholinesterase**, the enzyme that removes acetylcholine from synapses. Acetylcholine therefore accumulates at all three sites where it acts (Eddleston et al., 2008):
  - **muscarinic** → DUMBELS: Diarrhoea, Urination, Miosis, Bronchorrhoea, Bronchospasm and Bradycardia, Emesis, Lacrimation, Salivation and sweating;
  - **nicotinic** (neuromuscular junction) → the muscle is first overstimulated (**fasciculations**) and then exhausted (**weakness and paralysis**, including of the respiratory muscles);
  - **CNS** → agitation, seizures, coma and central respiratory depression.
- **Why the patient dies**: **respiratory failure**, from four things together — the airways fill with secretions (bronchorrhoea), the bronchi constrict (bronchospasm), the respiratory muscles fail, and the brainstem stops driving breathing (Eddleston et al., 2008).
- **"Ageing"**: over time the phosphorylated enzyme undergoes a chemical change ("ageing"), after which it is **permanently** inactivated. The body must make new enzyme, which is why poisoning can last for days. **Carbamates** bind **reversibly**, so their toxicity is usually shorter (Eddleston et al., 2008).
- **Why atropine works, and its limit**: atropine is a **competitive muscarinic antagonist**. It blocks the muscarinic effects — it **dries the secretions**, opens the bronchi and corrects the bradycardia. It **does nothing at nicotinic receptors**, so **it does not reverse muscle weakness**; the patient may still need **ventilation**.
- **Why you double the dose**: the amount of acetylcholine is enormous and unpredictable. Doubling every 3–5 minutes reaches an effective dose quickly. The end point is a **clear chest** (no crackles or wheeze), a heart rate above 80 and a systolic BP

above 80 — **not the pupil size** (Eddleston et al., 2008). The CDST uses 2 mg, doubled every 3 minutes, then 20% of the total dose per hour (HPCSA PBEC, 2025).

- **Why oximes are used** *(hospital)*: pralidoxime and obidoxime can **reactivate** the enzyme by removing the phosphate group, but only **before ageing** occurs. Their benefit in pesticide poisoning is uncertain (Eddleston et al., 2008).
- **Why no succinylcholine**: succinylcholine is broken down by **plasma cholinesterase**, which is also inhibited, so its paralysis lasts far longer than expected (Eddleston et al., 2008).
- **Why PPE and decontamination come first**: organophosphates are absorbed **through intact skin**, so the patient's clothing and vomit can poison you.
- **Say**: "*Acetylcholinesterase inhibition causes acetylcholine excess at muscarinic, nicotinic and CNS receptors; death is from respiratory failure. Atropine dries the chest but does not fix the weakness, so I titrate atropine to a dry chest and support ventilation.*"

C8. Carbon monoxide — "the oximeter lies"

- **Mechanism** (Rose et al., 2017):
  1. CO binds haemoglobin **about 200–250 times more strongly than oxygen**, forming carboxyhaemoglobin (COHb), so **less oxygen is carried**.
  2. It **shifts the oxyhaemoglobin dissociation curve to the left**, so the oxygen that *is* carried is **held on to** and not released to the tissues.
  3. It binds **myoglobin** (heart muscle dysfunction) and **cytochrome c oxidase** (the cells cannot use oxygen properly).
  4. It triggers inflammation and lipid peroxidation in the brain → **delayed neurological sequelae** weeks later.
- **Why SpO_2 is falsely normal**: a standard pulse oximeter uses only two wavelengths of light and **reads COHb as if it were oxyhaemoglobin** (Rose et al., 2017). **Ignore the SpO_2.**
- **Signs**: headache (the most common), nausea, dizziness and confusion; **several people in one house, in winter, with a brazier or generator**; in severe cases syncope, seizures, coma and myocardial ischaemia.
- **Why 100% oxygen works**: oxygen competes with CO for haemoglobin. The half-life of COHb is about 4–6 hours on room air, about 60–90 minutes on 100% oxygen, and about 20–30 minutes with hyperbaric oxygen (Rose et al., 2017).
- **Why the fetus is especially at risk**: fetal haemoglobin binds CO even more strongly, and fetal COHb levels clear more slowly (Rose et al., 2017).
- **Say**: "*CO binds haemoglobin 200 to 250 times more avidly than oxygen and shifts the curve to the left, so the tissues starve while the oximeter reads normal; 100% oxygen shortens its half-life from about five hours to about one.*"

C9. Cyanide — "oxygen everywhere, but the cells cannot use it"

- **Mechanism**: cyanide binds the ferric iron of **cytochrome c oxidase** (the final step of the mitochondrial electron transport chain) and **blocks aerobic respiration**. Oxygen is delivered to the cells, but they **cannot use it**, so they switch to anaerobic metabolism → **severe lactic acidosis**. The brain and heart, which have the highest oxygen demand, fail first (Henretig, Kirk and McKay, 2019; Borron et al., 2007).

- **Signs**: sudden onset of dyspnoea, confusion, seizures, coma and cardiovascular collapse, often in **house fires** (burning plastics and synthetic materials) and in mining and industry. The SpO_2 may be normal, because the oxygen is still bound to haemoglobin; it is simply not being used.
- **Why hydroxocobalamin works** *(hospital)*: hydroxocobalamin **binds cyanide to form cyanocobalamin (vitamin B12)**, which is harmless and excreted in the urine. It turns the skin and urine red (Borron et al., 2007).
- **Why nitrites are avoided in smoke inhalation**: nitrites work by creating methaemoglobin (which binds cyanide), but methaemoglobin cannot carry oxygen. A fire victim may already have carboxyhaemoglobin, so this would further reduce the oxygen-carrying capacity.
- **Say**: "*Cyanide blocks cytochrome oxidase, so cells cannot use oxygen and produce lactic acid; I give 100% oxygen and transport for hydroxocobalamin, which binds it as vitamin B12.*"

C10. Corrosives — acids vs alkalis

- **Alkalis** (drain cleaners, caustic soda) cause **liquefactive necrosis**: they **saponify fats** (turn them into soap) and dissolve proteins. The tissue liquefies and the alkali keeps **penetrating deeper** until it is neutralised. Alkalis are often tasteless, so larger amounts are swallowed, and they injure mainly the **oesophagus**, with a risk of perforation and later stricture.
- **Acids** (battery acid, toilet cleaners) cause **coagulative necrosis**: they denature proteins and form a firm **eschar** (scab) that **limits deeper penetration**. Acids are painful and bitter, so smaller amounts are swallowed; they pass quickly through the oesophagus and injure mainly the **stomach**, and systemic absorption causes metabolic acidosis.
- **Why there is no emesis**: vomiting brings the corrosive back up across the oesophagus a **second time** and risks aspiration.
- **Why there is no neutralisation**: the neutralisation reaction releases **heat** (it is exothermic), adding a thermal burn to the chemical one.
- **Why there is no charcoal**: it does not bind corrosives, and it obscures the view at endoscopy (HPCSA PBEC, 2025).
- **Why the airway is the priority**: swelling of the lips, tongue, epiglottis and larynx causes **drooling, stridor and voice change**, which predicts airway obstruction.
- **Say**: "*Alkalis liquefy and penetrate the oesophagus; acids coagulate and form an eschar but hit the stomach; I never induce vomiting or neutralise, because that re-burns the oesophagus and adds a thermal injury, and I watch the airway.*"

C11. Hydrofluoric acid (HF) — "pain out of proportion"

- **Mechanism**: HF is a **weak acid**, so it stays mostly **non-ionised** and **penetrates deep** through intact skin without much surface burn. In the deep tissues, it releases **fluoride ions**, which **bind calcium and magnesium** and form insoluble salts. This destroys cells and, if enough is absorbed, causes **systemic hypocalcaemia and hypomagnesaemia** → QT prolongation, **VF** and cardiac arrest (Tintinalli et al., 2020).

- **Why the pain is severe while the skin looks normal**: the low calcium makes nerves fire continuously, and the damage is deep, not on the surface.
- **Why calcium gluconate gel and IV calcium work**: you **supply calcium** for the fluoride to bind instead of the patient's own calcium, and **relief of the pain is the sign that it is working**.
- **Say**: "*Fluoride binds calcium and magnesium, causing deep tissue destruction and systemic hypocalcaemia with VF; I irrigate and give calcium, which binds the fluoride, and pain relief is my endpoint.*"

C12. Paraffin and other hydrocarbons — "the danger is in the lungs, not the stomach"

- **Mechanism** (Tintinalli et al., 2020): paraffin has **low viscosity** (it flows easily) and **low surface tension** (it spreads easily), so even a tiny amount **creeps into the airway** during swallowing, coughing or vomiting. In the alveoli it **dissolves surfactant**, so the alveoli collapse, causing ventilation–perfusion mismatch, hypoxia and **chemical pneumonitis**. Halogenated and aromatic hydrocarbons (the solvents used in inhalant abuse) also **sensitise the myocardium to catecholamines**, causing VF — "**sudden sniffing death**".
- **Signs**: coughing, choking and gagging at the time of ingestion (which shows that aspiration has occurred), then tachypnoea, wheeze, crackles, hypoxia and fever; symptoms may develop over about six hours.
- **Treatment logic**:
  - **No vomiting**, because a second passage through the throat means a second chance to aspirate.
  - **No charcoal**, because hydrocarbons are poorly adsorbed and the main risk is aspiration, not absorption.
  - **Oxygen**, and **observation for at least six hours**.
  - In inhalant abuse, **do not startle or chase** the patient, and avoid adrenaline unless they are in arrest, because a catecholamine surge can trigger VF.
- **Say**: "*Paraffin's low viscosity and surface tension make it spread into the lungs and destroy surfactant, so the danger is aspiration pneumonitis; I never induce vomiting or give charcoal, give oxygen and observe for at least six hours.*"

C13. Toxic alcohols — methanol and ethylene glycol

- **Mechanism**: the parent alcohols are only mildly toxic (they cause **drunkenness without the smell of alcohol**), but **alcohol dehydrogenase** converts them into toxic acids (Kraut and Mullins, 2018):
  - **Methanol** → formaldehyde → **formic acid**, which poisons the mitochondria of the **retina and optic nerve** → blurred vision, "snowstorm" vision and **permanent blindness**, plus **high anion gap metabolic acidosis**.
  - **Ethylene glycol** (antifreeze) → **glycolic acid** (the main cause of the acidosis) → **oxalic acid**, which combines with calcium to form **calcium oxalate crystals** that block the kidney tubules → **acute kidney injury**, and also **hypocalcaemia**.

- **Why symptoms are delayed**: the toxic acids take hours to form, so the patient may look only drunk at first.
- **Why fomepizole or ethanol works** *(hospital)*: fomepizole **blocks alcohol dehydrogenase**, so the toxic acids are not made. Ethanol works because alcohol dehydrogenase prefers ethanol, so it is kept busy and the toxic alcohol is not converted (competitive inhibition). **Dialysis** removes both the alcohol and its acids (Kraut and Mullins, 2018).
- **Why no charcoal**: alcohols are not adsorbed (HPCSA PBEC, 2025).
- **Say**: "*Alcohol dehydrogenase turns methanol into formic acid, which causes blindness, and ethylene glycol into glycolic and oxalic acid, which cause acidosis and kidney failure; fomepizole blocks that enzyme.*"

C14. Ethanol — "hypoglycaemia in children, hidden injuries in adults"

- **Mechanism**: ethanol **enhances GABA-A** and **inhibits NMDA**, causing CNS depression (Tintinalli et al., 2020). When the liver metabolises ethanol, the ratio of NADH to NAD^+ rises, which **inhibits gluconeogenesis** (the liver's production of new glucose).
- **Why children become hypoglycaemic**: children have **small glycogen stores**. Once these run out, the liver cannot make new glucose because ethanol has blocked gluconeogenesis, so even a small amount of alcohol can cause hypoglycaemic seizures in a child (Tintinalli et al., 2020).
- **Why thiamine**: thiamine is a cofactor for key enzymes of glucose metabolism (such as pyruvate dehydrogenase). In thiamine-deficient people (heavy drinkers and the malnourished), the brain cannot use glucose properly, leading to **Wernicke's encephalopathy** (confusion, ataxia and ophthalmoplegia). This is why the CDST gives thiamine 100 mg to people with alcohol misuse or malnutrition (HPCSA PBEC, 2025).
- **Why "drunk" is a diagnosis of exclusion**: alcohol masks head injury, hypoglycaemia, hypothermia (vasodilatation causes heat loss) and other ingestions.

C15. Opioids and naloxone

- **Mechanism**: **μ-opioid receptor** agonism → **reduced brainstem response to CO_2 and hypoxia** → slow, shallow breathing, then apnoea; miosis via the Edinger–Westphal nucleus; sedation (see the opioid row in Part B).
- **Why naloxone works**: it is a **competitive antagonist at the μ receptor**, so it displaces the opioid and reverses respiratory depression.
- **Why it is titrated and why the CDST starts at 0.04 mg in known opioid users** (HPCSA PBEC, 2025):
    1. In opioid-dependent people, a large dose suddenly removes all opioid effect, precipitating **acute withdrawal** — vomiting (with aspiration risk in a drowsy patient), agitation and violence.
    2. The aim is to **restore breathing, not full consciousness**.
- **Why patients can stop breathing again (re-narcotisation)**: naloxone's effect lasts a relatively short time, often shorter than that of the opioid (especially methadone or long-acting preparations), so the opioid can take over again once the naloxone wears off. This is why these patients are **monitored and transported**.

- **Say**: "*Naloxone competitively displaces the opioid from the μ receptor; I titrate it to adequate breathing, not to wakefulness, because a large dose precipitates withdrawal, and I transport because it can wear off before the opioid does.*"

C16. Beta-blockers, calcium channel blockers and digoxin-like plants

- **Beta-blockers**: they block **β_1 receptors** in the heart → **bradycardia, AV block and hypotension**, and β_2 receptors → bronchospasm and **hypoglycaemia** (β_2 stimulation normally drives glycogen breakdown — children are especially affected). **Propranolol** also blocks **sodium channels** → wide QRS and seizures (Tintinalli et al., 2020).
  - **Why glucagon works**: glucagon has **its own receptor** on heart cells, which increases cAMP **without needing the β receptor**, so it bypasses the blockade and increases heart rate and contractility (Lavonas et al., 2023).
- **Calcium channel blockers**: they block **L-type calcium channels** in the heart → bradycardia, AV block and weak contraction; in the blood vessels → **vasodilatation and hypotension**; and in the pancreas → **reduced insulin release, causing hyperglycaemia**, because the β-cell needs calcium entry to release insulin (Tintinalli et al., 2020).
  - **Why IV calcium works**: raising the calcium concentration outside the cell **forces more calcium through the channels that are still open** (Lavonas et al., 2023). **Adrenaline** supports heart rate and blood pressure when this is not enough.
- **Cardiac glycosides** (digoxin, oleander, slangkop, climbing onion and plakkie plants): they **inhibit the Na^+/K^+-ATPase pump**. Intracellular calcium rises (causing ectopy and arrhythmias), vagal tone increases (causing bradycardia and AV block), and potassium cannot be pumped back into cells, causing **hyperkalaemia**, which is a marker of severity in acute poisoning (Tintinalli et al., 2020).
  - **Why digoxin-specific Fab works** *(hospital)*: antibody fragments **bind the glycoside** and remove it from the pump.

C17. Stimulants (cocaine, methamphetamine "tik")

- **Mechanism**: amphetamines **release** noradrenaline, dopamine and serotonin and block their reuptake; cocaine **blocks their reuptake** and also **blocks sodium channels** (a local anaesthetic effect) → wide QRS and arrhythmias (Tintinalli et al., 2020).
- **Why chest pain**: noradrenaline causes **coronary vasoconstriction**, raises heart rate and blood pressure (increasing myocardial oxygen demand), and makes platelets more likely to clot. Oxygen supply falls while demand rises, causing ischaemia.
- **Why benzodiazepines are first-line** (the CDST gives diazepam or midazolam for stimulant chest pain): they reduce the **central sympathetic outflow** from the brain, which lowers heart rate, blood pressure, agitation and temperature together (HPCSA PBEC, 2025).
- **Why hyperthermia is the killer**: muscle activity plus vasoconstriction (which prevents heat loss) causes hyperthermia, then rhabdomyolysis, then acidosis, hyperkalaemia and arrest — the same pathway as ABD (A3). This is why the CDST instructs active cooling at 38 °C or above.

C18. Colchicine (and flame lily) and theophylline

- **Colchicine**: it binds tubulin and **stops microtubules forming**, so cells cannot divide. **Rapidly dividing cells fail first** — gut lining (severe vomiting and diarrhoea at 0–24 hours), bone marrow (pancytopenia and sepsis at days 2–7), and hair follicles (**alopecia** at about 2 weeks) — with multi-organ failure in between (Finkelstein et al., 2010).
- **Theophylline**:
    1. **Adenosine antagonism** → seizures and arrhythmias. Adenosine is the brain's natural anticonvulsant and the heart's natural "brake", which is why the seizures are so hard to stop.
    2. **Phosphodiesterase inhibition** → increased cAMP → tachycardia and vasodilatation.
    3. **Catecholamine release** → tachyarrhythmias and **hypokalaemia**, because β_2 stimulation drives potassium into cells.
    - **Intractable vomiting** is characteristic (Tintinalli et al., 2020). **Multiple-dose activated charcoal** enhances its elimination.

C19. Chlorine gas

- **Mechanism**: chlorine has **intermediate water solubility**, so it is not completely absorbed by the moist upper airway (as highly soluble ammonia would be), and it reaches the **lower airways and alveoli** as well. On moist surfaces it forms **hydrochloric acid and hypochlorous acid** and free radicals, which burn the airway lining and damage the alveolar–capillary membrane (Henretig, Kirk and McKay, 2019).
- **Signs**: immediate eye and throat irritation, cough, chest tightness, wheeze and stridor, then possible **delayed non-cardiogenic pulmonary oedema**, as the damaged capillaries leak over the following hours.
- **Treatment logic**:
    - Move **upwind**, to fresh air (chlorine is heavier than air and collects in low areas).
    - Give oxygen and **salbutamol** for bronchospasm.
    - **Observe and transport**, because pulmonary oedema can be delayed.

Part D — Decontamination: why it works and why it fails

D1. Activated charcoal

- **Mechanism**: activated charcoal is made with an **enormous internal surface area** of pores. Drug molecules in the gut **adsorb** (stick) to that surface by weak chemical forces, so they stay in the gut and pass out in the faeces instead of being absorbed into the blood (Tintinalli et al., 2020).
- **Why it must be early (within 1 hour)**: it can only bind drug that is **still in the gut lumen**; once the drug has been absorbed, single-dose charcoal cannot reach it.
- **Why it fails with certain poisons**: charcoal binds large, non-polar molecules well, but it binds poorly to **small, ionised or highly polar substances**:
  - **metals** (iron and lithium);
  - **alcohols** (ethanol, methanol and ethylene glycol);
  - **acids and alkalis**;
  - **hydrocarbons** — these are also excluded because of the risk of aspiration.

  These are exactly the CDST exclusions (HPCSA PBEC, 2025).
- **Why "fully conscious" is required**: a drowsy patient can **vomit and aspirate** charcoal into the lungs, which causes severe pneumonitis.
- **Why multiple-dose charcoal (MDAC) works for some drugs** *(hospital)*: some drugs (carbamazepine, phenobarbital, dapsone, quinine and theophylline) are **re-secreted into the gut** in bile or **diffuse back from the blood across the gut wall**. Repeated charcoal keeps the gut concentration at zero, which **pulls the drug out of the blood** — this is sometimes called "**gut dialysis**" — and interrupts the **enterohepatic circulation**.

D2. Why we do not induce vomiting

Ipecac-induced vomiting removes only a variable and often small proportion of the stomach contents, **has never been shown to improve outcome**, delays charcoal and oral antidotes, and causes **aspiration** (Höjer et al., 2013). The CDST says to avoid inducing vomiting (HPCSA PBEC, 2025).

D3. Skin and eye decontamination

Many poisons — **organophosphates** in particular — are absorbed **through intact skin**, and corrosives continue burning for as long as they are in contact with the skin. **Removing clothing** removes most of the contamination. **Washing with soap and water** removes the rest. **Irrigating the eyes for 20 minutes** dilutes the chemical and returns the pH towards normal (HPCSA PBEC, 2025).

Part E — Bites, stings, plants and food poisoning

E1. Snake venom — the three syndromes

Cytotoxic ("swell") — puff adder, Gaboon adder and Mozambique spitting cobra (Müller et al., 2012):

- **Mechanism**: **proteolytic enzymes, phospholipases and metalloproteinases** digest tissue and **damage the walls of blood vessels**. Plasma leaks out into the limb.
- **Signs**: immediate pain, **progressive swelling**, blistering, bruising and necrosis. **Hypovolaemic shock** can occur, because litres of fluid can move into a swollen limb.
- **Why no pressure immobilisation bandage (PIB)**: the bandage **traps the venom in the limb**, concentrating the tissue destruction and worsening necrosis.
- **Treatment logic**: IV fluids for shock, analgesia (no NSAIDs, because of the bleeding risk), splinting, marking the edge of the swelling, and antivenom when swelling is rapid or extensive.

Neurotoxic ("weak") — mambas and non-spitting cobras (Cape, snouted and forest):

- **Mechanism**: **Cobra α-neurotoxins** are **postsynaptic** — they bind the acetylcholine receptor on the muscle, so acetylcholine cannot act (a "curare-like" block). **Mamba venom** contains **dendrotoxins and fasciculins**, which increase acetylcholine release and block its breakdown, causing **fasciculations** before the muscle fatigues. Either way, **neuromuscular transmission fails** (Müller et al., 2012; Gutiérrez et al., 2017).
- **Signs**: the smallest and most active muscles fail first — **ptosis** and double vision, then **bulbar paralysis** (dysphagia, drooling and slurred speech), then **descending paralysis** to the **respiratory muscles**, causing death from respiratory failure, sometimes within 30 minutes to a few hours of a mamba bite.
- **Why PIB and keeping still help**: venom spreads mainly through the **lymphatics**, and lymph is pumped by **muscle movement**. Keeping the patient still and bandaging the limb slows the venom's spread to the circulation (Müller et al., 2012).
- **Why bag-valve-mask ventilation saves lives**: the venom paralyses the muscles but **does not damage the brain or heart**. If you breathe for the patient, they survive until antivenom or time reverses the block.

Haemotoxic ("bleed") — boomslang and vine snake:

- **Mechanism**: a **procoagulant** venom activates prothrombin, so the clotting factors are **used up** throughout the circulation. This is **venom-induced consumption coagulopathy**, and the blood becomes **unable to clot** (Müller et al., 2012).
- **Why it is delayed**: the clotting factors take hours to be consumed, so bleeding (from the gums, venepuncture sites and old wounds, haematuria and intracranial haemorrhage) may start **more than 24 hours** after a bite that looked trivial.
- **Treatment logic**: the boomslang requires its **own monovalent antivenom**, because the polyvalent antivenom does not cover it.

Antivenom:

- **Mechanism**: it is made from **horse antibodies** that **bind and neutralise venom still circulating** in the body. It **cannot repair tissue already destroyed** or reverse damage that has already happened, which is why it is given when indicated, not "just in case", and not delayed once indicated (Gutiérrez et al., 2017).
- **Why adrenaline must be ready**: these are **foreign (horse) proteins**, so early **anaphylactic reactions** are common. **Serum sickness** (an immune-complex reaction) can occur days later (Müller et al., 2012).

Spat venom in the eyes: the venom is directly cytotoxic to the cornea, so **immediate, copious irrigation** dilutes and removes it before it causes corneal ulceration and blindness.

E2. Spiders and scorpions

- **Button spider** (*Latrodectus*): its venom (α-latrotoxin) forms pores in nerve endings, causing a **massive release of neurotransmitters** — acetylcholine and noradrenaline (Isbister and Fan, 2011). This explains the **severe muscle pain and cramps**, the **rigid "board-like" abdomen** (which mimics an acute abdomen), **sweating**, hypertension and tachycardia.
- **Violin spider** (*Loxosceles*): its venom enzyme (**sphingomyelinase D**) damages cell membranes and blood vessels and triggers inflammation. Small vessels clot and the skin dies, producing a **necrotic ulcer** over days (Isbister and Fan, 2011).
- ***Parabuthus* scorpions** ("thick tail, thin pincers"): their venom **prolongs the opening of sodium channels** in nerves, so nerves keep firing and release excess neurotransmitters, both sympathetic and parasympathetic. This explains the **intense pain and paraesthesia**, **drooling**, sweating, dysphagia, muscle twitching and ptosis, and the **respiratory failure** in small children and the elderly, who must be ventilated.

E3. Marine envenomation and toxins

- **Why hot water works for stonefish, stingray and bluebottle stings**: these venoms are **heat-labile proteins**, which are **denatured (inactivated) by heat**. Immersion in hot water at about 45 °C relieves pain better than ice packs for bluebottle stings (Loten et al., 2006).
- **Tetrodotoxin** (pufferfish, "blaasop") and **saxitoxin** (paralytic shellfish poisoning in mussels from west coast **red tides**): both **block sodium channels in nerves**, so **no action potentials** are generated. This explains the perioral tingling and numbness, then **ascending paralysis and respiratory failure**, with the patient **often fully conscious**. Both toxins are **heat-stable**, so cooking does not help. The treatment is **ventilation** until the toxin is eliminated.

E4. Poisonous plants and traditional medicine

- ***Datura*** (thorn apple, "malpitte"): its **tropane alkaloids** (atropine and scopolamine) block **muscarinic receptors**, causing the full **anticholinergic toxidrome** (see Part B). Because anticholinergics **slow the gut**, absorption continues and the effects can last **24–48 hours**.
- ***Callilepis laureola*** (Impila): **atractyloside**-type compounds block the transporter that moves ADP into the mitochondria and ATP out (the **mitochondrial ADP/ATP carrier**). The cells run out of energy, and the organs with the highest metabolic demand

fail — **hepatic and renal tubular necrosis**. **Hypoglycaemia** occurs because the failing liver cannot release or make glucose (Popat et al., 2001).

- **Oleander and slangkop**: cardiac glycosides (see C16).
- ***Amanita phalloides*** (death cap mushroom): its **amatoxins** block **RNA polymerase II**, so liver cells can no longer make new proteins. The cells keep working on their existing proteins for a while, which is why the GI symptoms only start **more than 6 hours** after eating, followed by a false recovery, then **liver failure on days 2–5** (Tintinalli et al., 2020). **Say**: "*Late GI symptoms after mushrooms means amatoxin until proven otherwise.*"
- ***Dieffenbachia*** and arum lily: needle-like **calcium oxalate crystals** are injected into the mucosa when the plant is chewed, causing immediate burning pain and swelling of the lips and tongue.
- ***Nicotiana glauca*** (wild tobacco): its **anabasine** stimulates **nicotinic receptors**, first over-exciting the muscles (fasciculations) and then blocking them (paralysis) — like an organophosphate's nicotinic effects, without the muscarinic ones.

E5. Scombroid poisoning — "hot fish, hot face"

- **Mechanism**: dark-fleshed fish (tuna, mackerel and sardines) contain a lot of the amino acid **histidine**. If the fish is **not kept cold enough** after it is caught, **bacteria convert histidine into histamine** (bacterial decarboxylation). The fish may look and smell normal. **Histamine is heat-stable**, so cooking, canning or smoking kills the bacteria but **does not destroy the histamine already made** (Hungerford, 2010).
- **Why the signs look exactly like an allergic reaction**: the patient has swallowed a **large dose of histamine**.
  - Histamine acting on **H1 receptors** → vasodilatation (**flushing of the face and upper body**, headache and hypotension), urticaria, itch and bronchospasm.
  - Acting on **H2 receptors** → more vasodilatation and gastric effects (nausea, vomiting, abdominal cramps and diarrhoea).
  - A **peppery, metallic or burning taste** in the mouth is characteristic.
  - Onset is fast (**10–30 minutes**), because no immune reaction is needed — the histamine is already in the food (Hungerford, 2010).
- **Why it is not an allergy**: no **IgE antibodies** are involved, so the patient does not need to have been sensitised before. **Several people who ate the same fish are affected at the same time**, and the patient can eat the same fish safely in future if it is fresh. Mislabelling the patient as "fish-allergic" is a viva trap.
- **Why isoniazid makes it worse**: the gut normally breaks down swallowed histamine using enzymes such as **diamine oxidase**. **Isoniazid inhibits these enzymes**, so more histamine is absorbed (Hungerford, 2010). This is relevant in South Africa because many people take isoniazid for TB treatment or prevention.
- **Treatment logic**:
  - **H1 and H2 antihistamines** block the receptors the histamine acts on.
  - **Adrenaline** physiologically reverses vasodilatation and bronchospasm if there is hypotension or bronchospasm (the anaphylaxis-like picture).
  - **Oxygen, fluids and salbutamol** as needed.

    - The symptoms resolve within about **12–24 hours** as the histamine is metabolised.
    - **Keep the fish and notify environmental health**, because others may be at risk.
- **Say**: "*Bacteria in poorly chilled fish convert histidine into heat-stable histamine, so the patient has swallowed histamine. It looks like anaphylaxis but is not IgE-mediated and affects several diners; I treat with H1 and H2 blockers, and adrenaline if there is shock or bronchospasm.*"

Part F — Ten “why” questions to practise aloud

1. Why does a patient with *Datura* poisoning have dry skin, but a patient on “tik” is sweaty, when both have big pupils?
2. Why does a paracetamol overdose patient look well on day one, and why does NAC work?
3. Why must you never hypoventilate an intubated salicylate patient?
4. Why does sodium bicarbonate narrow the QRS in TCA toxicity, and why does the CDST tell you to hyperventilate after intubation?
5. Why does atropine dry the chest in organophosphate poisoning but not reverse the muscle weakness?
6. Why is the SpO_2 normal in carbon monoxide poisoning, and why does 100% oxygen help?
7. Why is charcoal useless for iron, lithium, alcohols, corrosives and paraffin?
8. Why does prone restraint with weight on the back cause sudden death in acute behavioural disturbance?
9. Why do we use a pressure bandage for a mamba bite but not for a puff adder bite?
10. Why is scombroid poisoning not an allergy, and why does isoniazid make it worse?

References for this section

American Geriatrics Society Beers Criteria Update Expert Panel (2023) 'American Geriatrics Society 2023 updated AGS Beers Criteria® for potentially inappropriate medication use in older adults', *Journal of the American Geriatrics Society*, 71(7), pp. 2052–2081.

American Psychiatric Association (APA) (2022) *Diagnostic and statistical manual of mental disorders, fifth edition, text revision (DSM-5-TR)*. Washington, DC: American Psychiatric Association Publishing.

Borron, S.W., Baud, F.J., Barriot, P., Imbert, M. and Bismuth, C. (2007) 'Prospective study of hydroxocobalamin for acute cyanide poisoning in smoke inhalation', *Annals of Emergency Medicine*, 49(6), pp. 794–801.

Boyer, E.W. and Shannon, M. (2005) 'The serotonin syndrome', *New England Journal of Medicine*, 352(11), pp. 1112–1120.

Chiew, A.L., Reith, D., Pomerleau, A., Wong, A., Isoardi, K.Z., Soderstrom, J. and Buckley, N.A. (2020) 'Updated guidelines for the management of paracetamol poisoning in Australia and New Zealand', *Medical Journal of Australia*, 212(4), pp. 175–183.

Decker, B.S., Goldfarb, D.S., Dargan, P.I., et al. (EXTRIP Workgroup) (2015) 'Extracorporeal treatment for lithium poisoning: systematic review and recommendations from the EXTRIP workgroup', *Clinical Journal of the American Society of Nephrology*, 10(5), pp. 875–887.

Eddleston, M., Buckley, N.A., Eyer, P. and Dawson, A.H. (2008) 'Management of acute organophosphorus pesticide poisoning', *The Lancet*, 371(9612), pp. 597–607.

Finkelstein, Y., Aks, S.E., Hutson, J.R., et al. (2010) 'Colchicine poisoning: the dark side of an ancient drug', *Clinical Toxicology*, 48(5), pp. 407–414.

Gutiérrez, J.M., Calvete, J.J., Habib, A.G., Harrison, R.A., Williams, D.J. and Warrell, D.A. (2017) 'Snakebite envenoming', *Nature Reviews Disease Primers*, 3, 17063.

Health Professions Council of South Africa, Professional Board for Emergency Care (HPCSA PBEC) (2025) *Clinical Decision Support Tools for Emergency Care Providers, South Africa 2024. Version 1.3, October 2025*. Pretoria: HPCSA.

Henretig, F.M., Kirk, M.A. and McKay, C.A. (2019) 'Hazardous chemical emergencies and poisonings', *New England Journal of Medicine*, 380(17), pp. 1638–1655.

Höjer, J., Troutman, W.G., Hoppu, K., et al. (2013) 'Position paper update: ipecac syrup for gastrointestinal decontamination', *Clinical Toxicology*, 51(3), pp. 134–139.

Hungerford, J.M. (2010) 'Scombroid poisoning: a review', *Toxicon*, 56(2), pp. 231–243.

Hüser, C., Bethlehem, C., Dünser, M.W., et al. on behalf of the ESICM Trauma, Emergency Medicine Section (2025) 'Critical care management of acute poisoning' (document supplied for this module).

Inouye, S.K., Westendorp, R.G.J. and Saczynski, J.S. (2014) ‘Delirium in elderly people’, *The Lancet*, 383(9920), pp. 911–922.

Isbister, G.K. and Fan, H.W. (2011) ‘Spider bite’, *The Lancet*, 378(9808), pp. 2039–2047.

Juurlink, D.N., Gosselin, S., Kielstein, J.T., et al. (EXTRIP Workgroup) (2015) ‘Extracorporeal treatment for salicylate poisoning: systematic review and recommendations from the EXTRIP workgroup’, *Annals of Emergency Medicine*, 66(2), pp. 165–181.

Kraut, J.A. and Mullins, M.E. (2018) ‘Toxic alcohols’, *New England Journal of Medicine*, 378(3), pp. 270–280.

Lavonas, E.J., Akpunonu, P.D., Arens, A.M., et al. (2023) ‘2023 American Heart Association focused update on the management of patients with cardiac arrest or life-threatening toxicity due to poisoning’, *Circulation*, 148(16), pp. e149–e184.

Loten, C., Stokes, B., Worsley, D., Seymour, J.E., Jiang, S. and Isbister, G.K. (2006) ‘A randomised controlled trial of hot water (45 °C) immersion versus ice packs for pain relief in bluebottle stings’, *Medical Journal of Australia*, 184(7), pp. 329–333.

McCutcheon, R.A., Reis Marques, T. and Howes, O.D. (2020) ‘Schizophrenia — an overview’, *JAMA Psychiatry*, 77(2), pp. 201–210.

Müller, G.J., Modler, H., Wium, C.A., Veale, D.J.H. and Marks, C.J. (2012) ‘Snake bite in southern Africa: diagnosis and management’, *Continuing Medical Education*, 30(10), pp. 362–381.

Popat, A., Shear, N.H., Malkiewicz, I., et al. (2001) ‘The toxicity of *Callilepis laureola*, a South African traditional herbal medicine’, *Clinical Biochemistry*, 34(3), pp. 229–236.

Ritter, J.M., Flower, R.J., Henderson, G., Loke, Y.K., MacEwan, D., Robinson, E. and Fullerton, J. (2023) *Rang & Dale’s pharmacology*. 10th edn. Edinburgh: Elsevier.

Rose, J.J., Wang, L., Xu, Q., et al. (2017) ‘Carbon monoxide poisoning: pathogenesis, management, and future directions of therapy’, *American Journal of Respiratory and Critical Care Medicine*, 195(5), pp. 596–606.

Royal College of Emergency Medicine (RCEM) (2025) *Acute behavioural disturbance in emergency departments. Best Practice Guideline, Version 3 (May 2025)*. London: RCEM.

Schuckit, M.A. (2014) ‘Recognition and management of withdrawal delirium (delirium tremens)’, *New England Journal of Medicine*, 371(22), pp. 2109–2113.

Tintinalli, J.E., Ma, O.J., Yealy, D.M., Meckler, G.D., Stapczynski, J.S., Cline, D.M. and Thomas, S.H. (eds) (2020) *Tintinalli’s emergency medicine: a comprehensive study guide*. 9th edn. New York: McGraw-Hill Education.